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Single Tree Segmentation and Diameter at Breast Height Estimation With Mobile LiDAR

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ABSTRACT The tree diameter at breast height (DBH) is one of the most important variables for monitoring the forest ecology. Mobile laser scanning (MLS), which has been widely applied in the forestry field, makes DBH measurement fast and convenient. However, there are many shrubs and deadwood in the neutral forest environment and the point clouds quality from MLS are easily affected by the environment which results in low single tree segmentation and DBH estimation accuracy. To improve the accuracy in a complex forest environment and low point cloud quality, we propose a relative density segmentation method for the single tree segmentation and DBH estimation method based on multi-height diameters for the DBH estimation. The relative density segmentation method calculates the relative density according to the ratio of density in two different scales, and segments the tree trunks by the higher relative density of trunk point clouds compared with their surroundings points. In the natural forest plot, the precision and recall of the proposed segmentation method reached 0.966 and 0.946, respectively; In the urban forest plot, the precision and recall reached 1 and 0.966, respectively. The proposed DBH estimation method was used to estimate the DBH of trees using multi-height diameters. The multi-height diameters combined with the outlier detection algorithm were able to improve the accuracy and robustness when the trunk point clouds have large noise. For the DBH estimation results, the mean absolute error (MAE), mean absolute percentage error (MAPE), and root mean square error (RMSE) were 2.5 cm, 11.54%, and 3.17 cm, respectively, in the natural forest plot and 1.65 cm, 6.31%, and 1.97 cm, respectively, in the urban forest plot. The good experiment results indicate that the proposed method can achieve accurate and robust DBH extraction and provide fundamental data for supervision and sustainable development of forest resources.

INDEX TERMS Diameter at breast height, mobile laser scanning, point cloud, single tree segmentation.

I. INTRODUCTION

The tree diameter at breast height (DBH) is the diameter of the trunk at 1.3m height above the ground surface [1]. Being able to obtain the tree DBH quickly is important for monitoring a forest ecosystem. Most traditional methods for finding the DBH are based on manual measurement, which require a great deal of manpower and working time. It is also difficult to collect periodic data in a large area within a short time.

In recent years, with the increasing application of Light Detection and Ranging (LiDAR) technology in the field

of forestry, lidar became one of the most important methods for forestry investigation [2]–[5]. Terrestrial laser scanning (TLS) and mobile laser scanning (MLS) are the most common ways of acquiring forest understory three-dimensional (3D) data [6]. TLS can achieve a higher scanning accuracy for the tree trunk and extract high-precision laser point cloud data.

Trochta *et al.* [7] segmented trees by the distance of the points-based approach and estimated the DBH of trees based on the point cloud obtained by TLS and least square regression (LSR), within an error of less than 1 cm compared with the measured value. Qi *et al.* [8] achieved an automatic estimation of DBH based on TLS data by using the ant colony algorithm and B-spline curve fitting algorithm.

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The root-mean-square error (RMSE) of the algorithm for extracting DBH was ± 0.19 cm, which was improved by 50% compared with the traditional algorithm. However, the equipment of TLS is generally huge, heavy, and relatively expensive. The TLS point cloud data is prone to occlusion and requires multiple stations to scan for complete scene data.

With the development of MLS and simultaneous localization and mapping (SLAM) technology [9], [10], the acquisition of forest 3D data became more convenient and efficient. The combination of MLS and SLAM algorithms enables real-time data collection and point cloud registration. Compared with the TLS, the MLS is easier and more convenient, because there is no need to set up a scanning site. It can be carried by hand or backpack to collect data by walking through the forest, which will save on post-processing time greatly [11].

Forsman *et al.* [12] used a two-dimensional (2D) mobile laser scanner and an inertial navigation system to obtain a 3D point cloud of the forest, using the point cloud of the trunk for the circle fit. The best result of RMSE at 14% (37 mm) was acquired with the circle fit on multiple 2D projected clusters. Zhou *et al.* [13] used a Velodyne 16 channel LiDAR sensor to obtain a forest 3D point cloud, segmented the trees using the Euclidean clustering algorithm, and used the random sample consensus (RANSAC) algorithm to circle fit the DBH of the trees in the plane. Xiao *et al.* [14] proposed a hierarchical clustering method for segmentation, a group of cross-section slices of the point cloud were formed at different heights, and the segmentation process of each slice was performed by cluster analysis. The random sample consensus algorithm was used to circle fit the point cloud slices to achieve a faster estimation of the tree DBH.

The correct single tree segmentation directly influences the DBH estimation and forest inventory. Scholars previously used the Euclidean cluster extraction method [13], [15], the hierarchical clustering method [14], and arc detection algorithms [16] etc. to segment the trees from the MLS point cloud, and some scholars developed a graph based segmentation for the Airborne Laser Scanning data [17], they all achieved good results. However in some forests, the forest ground was characterized by rugged terrain and obstacles, such as rocks, shrubs, and deadwood [18], and there are lots of shrubs and leaves that may influence the single tree segmentation. Meanwhile, because the backpack LiDAR obtains the forest point cloud by walking in the forest, the tree trunk point cloud density is higher than the surroundings, the density of the forest point cloud also relates to the distance from the LiDAR sensors, which leads to a different average density of the tree point cloud at different positions. In this study, we propose the relative density segmentation method for single tree segmentation. the relative density segmentation method segments the tree trunks and removes the shrubs and leaves by the higher trunks' point cloud relative density than shrubs and leaves. The relative density is calculated by the ratio of the point cloud density in two different scales

and, therefore, can be used to segment the tree trunk under different point cloud densities.

For the DBH estimation, scholars estimated DBH by fitting point cloud slice at 1.3 m with circle [12], [13], [19], [20] or fitting the tree trunk point cloud with cylinder [21], [22]. Because some tree trunk diameter has a lot of changes with different height, the cylinder may not fit to point cloud perfectly. Meanwhile, most MLS systems are backpack or hand-held mobile LiDAR system, and as mobile LiDAR sensors support 16 or fewer channels, the point cloud of MLS is relatively low accuracy, and the MLS point clouds quality are easily affected by walking path and walking speed, etc. The DBH estimation methods, such as circle fitting with the point cloud at 1.3 m height with different algorithms [11], [14], [20], [23], [24] may cause large error when the trunk point cloud exists with a great deal of noise. We propose a DBH estimation method based on multi-height diameters for the MLS point cloud DBH estimation. The proposed DBH estimation method estimates the tree trunk DBH by the multi-height diameters and uses different height diameters and the multi-height diameter outlier detection results to maintain the DBH estimation accuracy at a lower point cloud quality or with large noise.

This paper is organized as follows: The materials and methods are introduced in Section 2. The results and discussion are given in Section 3 and Section 4, followed by the conclusion in Section 5.

II. MATERIALS AND METHODS

A. BACKPACK LiDAR SYSTEM

The backpack mobile LiDAR system we used in this paper consisted of a LiDAR sensor, a micro-computer, a battery, an inertial measurement unit (IMU), and a remote-control client. The LiDAR sensor is Velodyne VLP-16 [25]. VLP-16 LiDAR supports 16 channels, the horizontal field of view is 360 degrees, the vertical field of view is ± 15 degrees, the data collection frequency is 10 HZ, the maximum measurement range is 100 meters, and the typical accuracy is ± 3 cm. The IMU is used to obtain the motion posture and movement of the system for the 3D SLAM algorithm. The micro-computer runs a 3D SLAM algorithm called Laser Odometry and Mapping (LOAM) [10] and drives the LiDAR sensor and IMU. The remote-control client connects with the micro-computer over the network and receives real-time data from the micro-computer. The backpack mobile LiDAR system component is shown in Fig.1.

B. EXPERIMENTAL DATA

We selected two different types of forests as the experiment forests, including natural forest and urban forest. The data were collected by walking in the forest with the backpack mobile LiDAR system, and the LOAM [10] algorithm was used to generate the forest point cloud.

The natural forest plot was a mixed (needle leaf and broadleaf) forest located in Erguna city, Inner Mongolia province, China. The size was approximately 68m * 52m

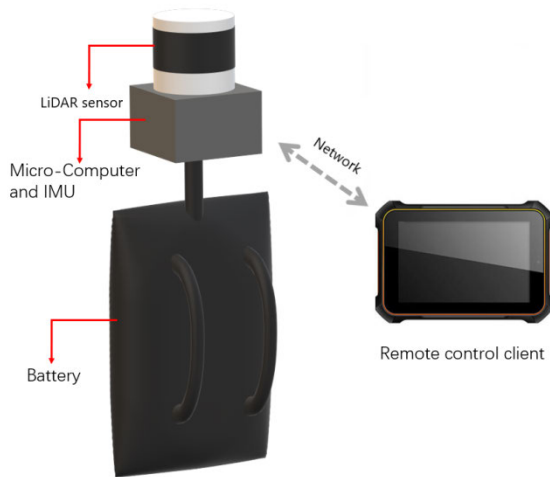


FIGURE 1. The backpack mobile LiDAR system components. Light Detection and Ranging (LiDAR), inertial measurement unit (IMU).

and with 148 trees in the plot, the average DBH of the trees was 24.6 cm, the maximum DBH was 40.5 cm, and the minimum DBH was 11.7 cm. The image of the natural forest plot is shown in Fig. 2 (a), and the natural forest ground is sloping and uneven, with a large amount of shrub and grass on the forest ground. The natural forest point clouds are shown in Fig. 2 (b) and 2(c). The point clouds of the trees are relatively noisier and in a lower quality.

The urban forest plot is an artificial park, located in Haidian District, Beijing, China. The size was approximately 44m * 48m and with 58 trees in the plot, the average DBH of the

trees was 31 cm, the maximum DBH was 62.4 cm, and the minimum DBH was 14.3 cm. The image of the urban forest plot is shown in Fig. 2 (d). The ground is relatively flat, and there are few shrubs and grasses on the forest ground. The urban forest point cloud is shown in Fig. 2 (e) and 2(f). The point cloud of trees has little noise and in a relatively simple environment.

C. METHODS

In this study, we remove the ground point cloud and normalize the off-ground point cloud to set the bottom of the trees at the same height. And we propose a relative density segmentation method for single tree trunk segmentation. The relative density segmentation method calculates the relative density by the ratio of point cloud density on different scales and extracts the tree trunks according to the relative density.

For the DBH estimation, the trunk point cloud quality may vary in different environments, and the DBH estimation methods, which use a certain height point cloud of the trunk to fit a circle or cylinder [11], [17], [18], [20], may produce a low accuracy when the trunk has large noise. Thus, we propose a DBH estimation method based on multi-height diameters. In this method, the RANSAC circle fit method is used to obtain the diameters at different heights of the tree trunk, and the outlier detection method is selected to detect and remove the outliers in multi-height diameters. The DBH value can be calculated by the multi-height diameters after removing the outliers. The flow chart of the proposed method is shown in Fig. 3.

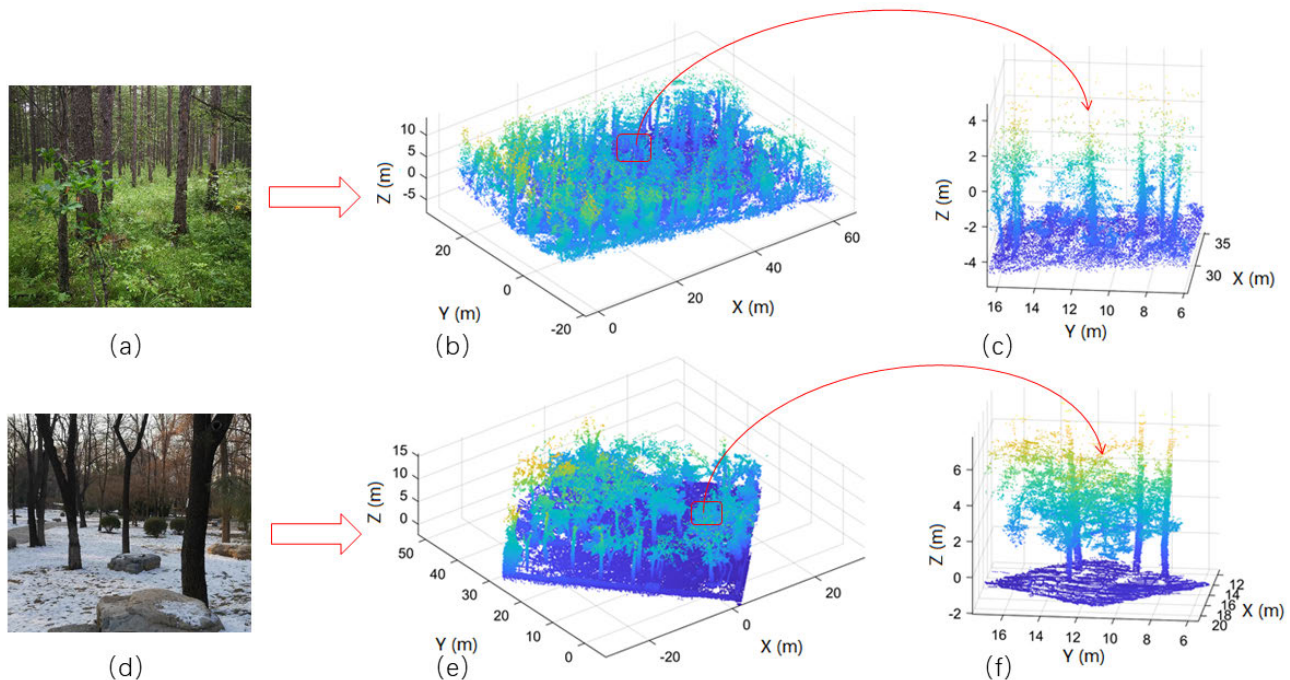


FIGURE 2. The forest images and point clouds. (a) The image of the natural forest plot; (b) the point cloud of the natural forest plot; (c) the point cloud of trees in the natural forest plot; (d) the image of the urban forest plot; (e) the point cloud of the urban forest plot; and (f) the point cloud of trees in the urban forest plot.

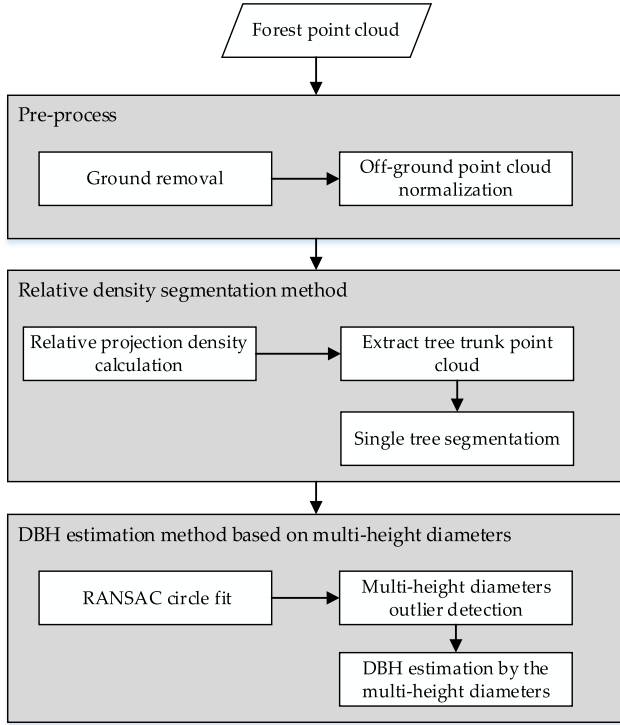


FIGURE 3. Flow chart of the proposed method. Tree diameter at breast height (DBH). random sample consensus (RANSAC).

1) PRE-PROCESS

Most natural forests are rugged and have sloping grounds, meaning the bottoms of trees are not on the same level, which may cause problems on segmentation and DBH estimation. This study removes the ground point cloud and corrects the off-ground forest point cloud to the same height according to the corresponding ground point cloud. This study selected cloth simulation filtering (CSF) [26] to remove the ground point cloud, the CSF is based on cloth simulation and is a common method for separating ground point clouds. As shown in Fig. 4(a) and 4(b), the ground point cloud was separated from the forest point cloud by the CSF.

As shown in Fig. 4(b), the ground point cloud had a slope, therefore, the bottoms of the trees were not on the same level. The Digital Elevation Model (DEM) generated by the ground point cloud was shown in Fig. 4(c), for each point cloud $p = (x_p, y_p, z_p)$ in the off-ground point cloud, the corrected point cloud $p' = (x_{p'}, y_{p'}, z_{p'})$ can be calculated by the following equation:

$$\begin{cases} x_{p'} = x_p \\ y_{p'} = y_p \\ z_{p'} = z_p - z_{pDEM} \end{cases} \quad (1)$$

where z_{pDEM} is the pixel value of p_{DEM} , and the p_{DEM} is the nearest pixel in DEM to the forest point p . The off-ground point cloud after normalization is shown in Fig. 4(d).

2) RELATIVE DENSITY SEGMENTATION METHOD

Point cloud segmentation is to segment the point cloud of a single tree from the whole forest point cloud, it is necessary

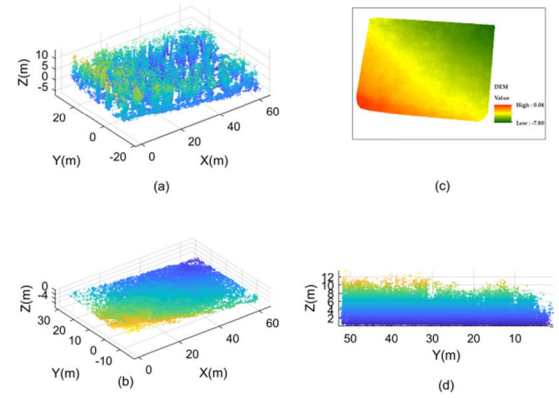


FIGURE 4. Ground point cloud removal and normalization. (a) Off-ground point cloud; (b) ground point cloud; (c) DEM generated by ground point cloud; and (d) off-ground point cloud after normalization.

for DBH estimation. Scholars previously used the hierarchical clustering method [14] to segment and locate the trees for DBH estimation, the hierarchical clustering method gets the cross-section slices of point cloud at a certain height and the segment the slices by cluster analysis. However, different from indoor scenarios or road scenarios, the forest environment is more complex. In dense forests, the shrubs and branch point clouds connect with the tree trunk point cloud, which may have an impact on single tree segmentation and lead to a lower segmentation accuracy. We proposed the relative density segmentation method to segment the trunk and remove the shrubs and branches.

The point cloud density can be used to extract the tree trunks, and scholars have used point cloud density to perform the segmentation of landslides, buildings, forests, etc. [27], [28]. As the backpack LiDAR obtains the forest point cloud by walking in the forest, the density of tree trunk point cloud after projected to XOY plane would be larger than shrubs and branches. The tree trunk can be extracted by setting a threshold to the point cloud density value, and there are some scholars who used the point density-based criterion to remove the trunk noise points. However, due to the non-uniformity of the forest point cloud, the point cloud density is related to the distance from the LiDAR sensor, and the varying point cloud density will affect the segmentation accuracy.

We propose a relative density segmentation method that uses the number of points in two different ranges to calculate the point cloud density projected to the XOY plane in different scales, as shown in Fig. 5(b), and the relative density can be calculated by the ratio of two different scales point cloud densities. As the tree trunk has a higher relative density than the surroundings, the tree trunk point clouds were extracted using the point cloud relative density. Finally, the point cloud segmentation method was used to segment the tree trunk point cloud into single trees.

As shown in Fig. 5, for the point $p = (x_p, y_p, z_p)$ in the normalized off-ground point cloud P , the projection distances $D_p = (d_1, d_2, \dots, d_n)$ are the distances on the XOY plane of

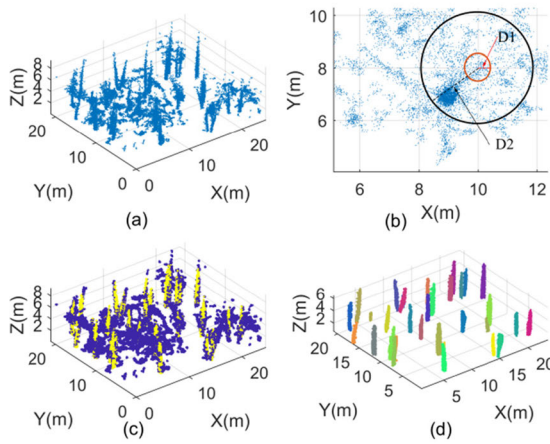


FIGURE 5. The relative density segmentation method. (a) The Off-ground point cloud after normalization, (b) the point cloud relative density calculating method, (c) the tree trunk point cloud extraction result, and (d) the single tree trunks.

point p to other points $p_i = (x_{p_i}, y_{p_i}, z_{p_i}) \in P$, ($1 \leq i < n$), and n is the point number of the point cloud P . The distance d_i can be expressed as:

$$d_i = \sqrt{(x_p - x_{p_i})^2 + (y_p - y_{p_i})^2}, \quad (i \in [1, n - 1]). \quad (2)$$

As shown in Fig. 5(b), we set the distance value D_1 and D_2 of two different scales, and the relative density value R_p of the point p can be calculated by the following equation:

$$R_p = \frac{N_{D1}}{N_{D2}} \quad (3)$$

where N_{D1} is the number of values in D_p that are smaller than D_1 , and N_{D2} is the number of values in D_p that are smaller than D_2 .

$$N_{D1} = \text{Count}(d_i < D_1), \quad (i \in [1, n - 1]). \quad (4)$$

$$N_{D2} = \text{count}(d_i < D_2), \quad (i \in [1, n - 1]). \quad (5)$$

D_2 is greater than D_1 , the relative density value R_p can represent the density of the location in point p relative to the surrounding. By setting a threshold to R_p , the trunk point cloud Q_{trunk} can be extracted. As shown in Fig. 5(c).

For the segmentation, as shown in Fig. 5(c), the tree trunks are separated from each other, and there are no branches or shrub point clouds that may influence the segmentation. Thus, we selected the Euclidean clustering method to segment the tree trunk point clouds. The segmented single tree trunks are shown in Fig. 5(d).

3) DBH ESTIMATION METHOD BASED ON MULTI-HEIGHT DIAMETERS

As shown in Fig. 6(a) and 6(b), the cross-section of the tree trunk appears as a circle, and the tree diameter at different heights can be obtained by the circle fitting of the trunk point cloud. The height of the tree's DBH is defined as 1.3 meters above the ground in most countries [1]. Therefore, most scholars estimate the DBH by slicing the point cloud at the height of 1.3 m and projecting it onto the XOY plane for circle

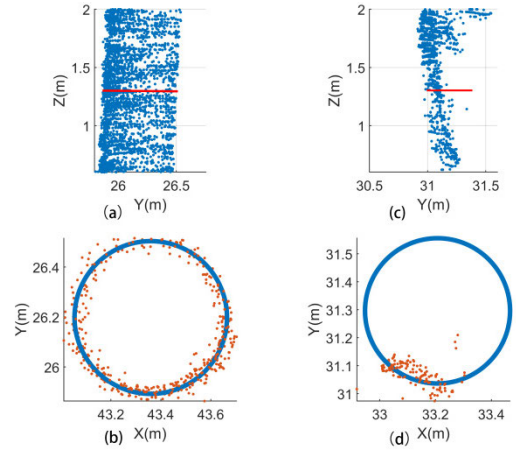


FIGURE 6. Trunk point cloud circle fitting with a 1.3 m height point cloud. (a) and (b) The circle fitting results of the point cloud at 1.3 m when the point cloud quality is good, 6(c) and (d) the circle fitting results of the point cloud at 1.3 m when the trunk point clouds have extra noise.

fitting [14], [23]. However, most point clouds of MLS have noise, and the segmented tree trunk point clouds may include some extra branches and shrub point cloud, which makes the result of point cloud circle fitting at 1.3 m inaccurate. As shown in Fig. 6(c) and Fig. 6(d), the tree trunk diameter obtained by the point cloud slice circle fitting at 1.3 m height is quite different from the measured value.

The proposed method obtains multi-height diameters by the circle fitting algorithm based on multi-height point clouds. The RANSAC [29] is selected as the circle fitting method in this study. RANSAC is one of the most used methods for curve fitting. This method can eliminate the influence of radius on fitting and work normally under a large curvature radius and certain noise.

The multi-height trunk point cloud circle fitting results are shown in Fig. 7. The tree trunk point cloud was selected from 0.7 m to 1.9 m and sliced with a thickness of 0.2 m, and the slice point cloud was projected onto the XOY plane for circle fitting. Seven diameters $D = (d_1, d_2, \dots, d_7)$ were obtained by circle fitting with the center heights $H = [0.7, 0.9, 1.1, 1.3, 1.5, 1.7, \text{ and } 1.9]$. However, as shown in Fig. 7(c) and 7(d), some circles at certain heights are quite different from the actual diameter, and these are called outliers. This study used the PauTa Criterion [30] to detect and remove the outliers. By calculating the multi-height trunk diameters and the outliers, the DBH can be calculated by the following equation:

$$\begin{cases} DBH = d_4, & (e_4 = 0) \\ DBH = \frac{1}{7 - n} \left(\sum_{k=1}^7 d_k - \sum_{i=1}^n e_i \right), & (n \in [0, 6], e_4 \neq 0) \end{cases} \quad (6)$$

where the $E = (e_1, e_2, \dots, e_7)$ is the diameter value of outliers, and n indicates the number of outliers. for $d_k \in D$, if $|d_k - \bar{D}| > 3\sigma$, then d_k is the outlier, and $e_k = d_k$, ($k \in [1, 7]$), if d_k is not the outlier, and $e_k = 0$, where $\bar{D}$ is the

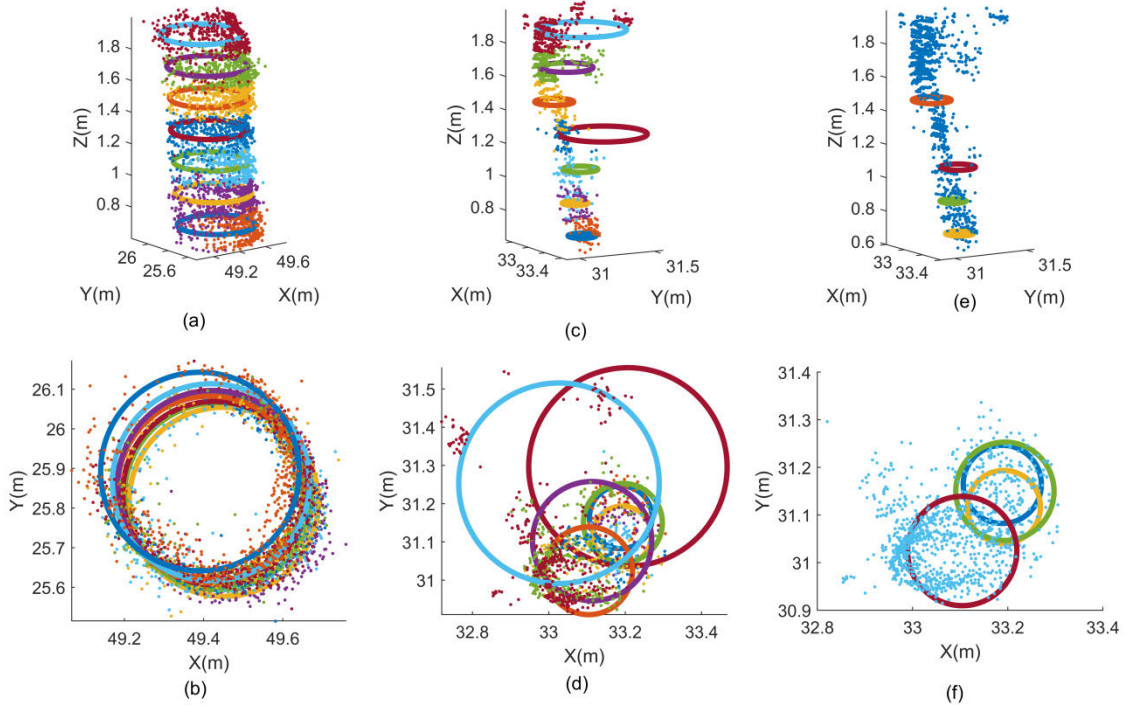


FIGURE 7. Trunk point cloud circle fitting with the multi-height trunk point cloud. (a) and (b) The circle fitting of a multi-height point cloud when the point cloud quality is good, (c) and (d) the circle fitting of a multi-height point cloud when the point cloud quality is poor, (e) and (f) the multi-height point cloud fitting circles after removing outliers.

average value of D , and σ is the standard deviation value of D .

For a trunk point cloud in good quality and have little extra point cloud that does not belong to the trunk, as shown in Fig. 7(a) and 7(b), there are no outliers in the multi-height diameters, the tree DBH is equal to d_4 , which means the diameter of the trunk in 1.3m height. For a tree trunk has lots of extra point cloud that does not belong to the trunk, as shown in Fig. 7(c), 7(d), the diameters of d_4 , d_6 , and d_7 are the outliers, as shown in Fig. 7(e), 7(f), we remove the outliers and the DBH can be calculated by the average value of rest diameters to get a closer value to the reference DBH.

4) ACCURACY OF TREE SEGMENTATION AND DBH ESTIMATION

In this study, precision and recall [31] were selected as the evaluation indexes of the point cloud segmentation method. Precision refers to the correct proportion of the segmentation result, and recall refers to the proportion of all correctly segmented targets, they were calculated as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

where the TP indicates the correctly segmented tree trunk point cloud, FP refers to the other point cloud in the plot that is wrongly segmented into the trunk point cloud, and FN refers to the missed tree trunk point cloud

The mean absolute error (MAE), mean absolute percentage error (MAPE) and root mean square error (RMSE) were

selected as the evaluation indexes for the DBH Estimation method, the MAE is used to measure the average absolute error between the estimated DBH dbh_i and the measured DBH dbh'_i , MAPE is used to measure the relative error between the estimated DBH dbh_i and the measured DBH dbh'_i , RMSE is used to measure the average quadratic deviation between the estimated DBH dbh_i and the measured DBH dbh'_i . They were calculated as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |dbh_i - dbh'_i| \quad (9)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \frac{|dbh_i - dbh'_i|}{dbh'_i} \times 100\% \quad (10)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n |dbh_i - dbh'_i|^2} \quad (11)$$

III. RESULTS

A. SINGLE TREE SEGMENTATION

To compare the accuracy of the proposed method, the hierarchical clustering method commonly used in single tree segmentation [13], [14] was selected as the comparison method, and we selected the point cloud at 1.3m height and 0.3m thick for the hierarchical clustering. And for the proposed segmentation method, we selected D_1 from [0.1, 0.2, 0.3, 0.4, 0.5] m and D_2 from [1, 2, 3, 4] m., for the different combinations of D_1 and D_2 , we calculated the corresponding relative density value R_p to find the best parameters for our experimental data. After comparison, when $D_1 = 0.3m$ and $D_2 = 3m$,

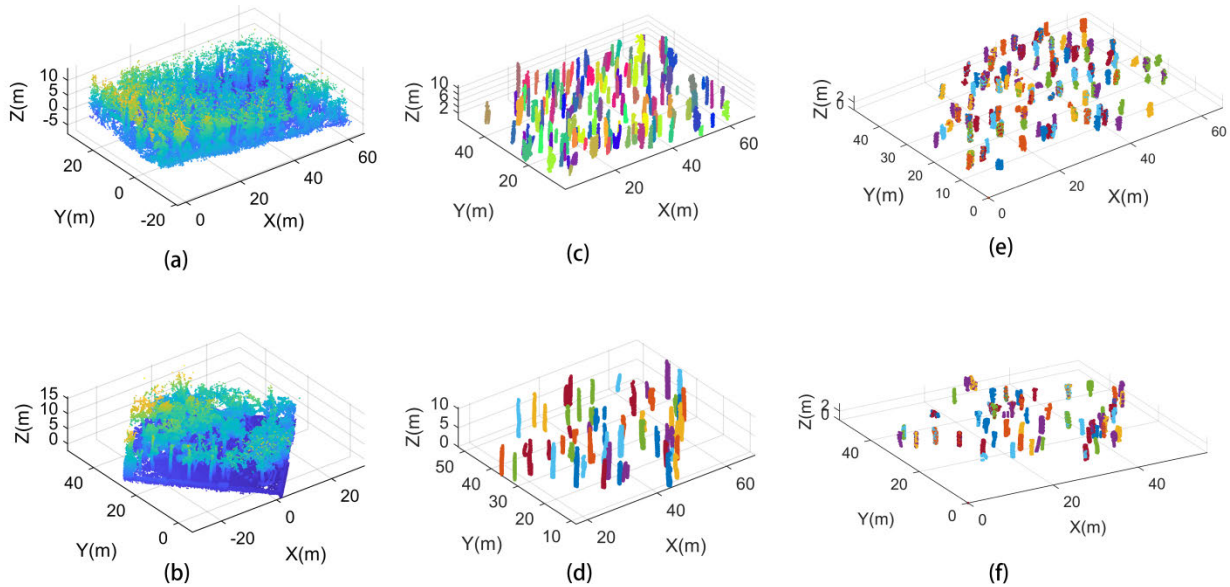


FIGURE 8. Comparison of the segmentation results of the proposed method and the hierarchical clustering method in the natural forest and urban forest. (a) The point cloud of the natural forest; (b) the point cloud of the urban forest; (c) the segmentation of the proposed method in the natural forest; (d) the segmentation of the proposed method in the urban forest; (e) the segmentation of the proposed method in the natural forest; and (f) the segmentation of the comparison method in the urban forest.

TABLE 1. The single tree segmentation results of the proposed method and the hierarchical clustering method.

Plots	Truth	Proposed method			Hierarchical clustering method		
		Correct	Wrong	Missed	Correct	Wrong	Missed
Natural forest	148	140	5	8	94	13	54
Urban forest	58	56	0	2	50	7	8

the tree trunk can be clearly distinguished from the other point cloud, by setting the threshold to 0.035 the trunk point cloud can be segmented. The single tree segmentation result is shown in Table 1 and Fig. 8. Compared with the truth data that manually labeled and counted, in the natural forest plot, 145 trees were extracted by the proposed segmentation method, and 140 of them were correct, 5 were wrong, and 8 were missed; the trees are shown in Fig. 8(c). We found 107 trees were extracted by the comparison method, 94 of them were correct, 13 were wrong, and 54 were missed; the trees are shown in Fig. 8(e). In the urban forest plot, 56 trees were extracted using the proposed method, and 56 trees were correct, 0 were wrong, and 2 were missed; the trees are shown in Fig. 8(d). Using the comparison method, 57 trees were extracted, and 50 trees were correct, 7 were wrong, and 8 were missed; the trees are shown in Fig. 8(f).

The precision and recall results are shown in Table 2. In the natural forest plot, the precision and recall of the proposed method were 0.966 and 0.946, respectively. The precision and recall of the hierarchical clustering method were 0.879 and 0.635, respectively. In the urban forest plot, the precision and recall of the proposed method were 1.00 and

TABLE 2. The precision and recall of the proposed method and the hierarchical clustering method.

Methods	Natural Forest		Urban Forest	
	precision	recall	precision	recall
Proposed Method	0.967	0.935	1.000	0.966
Hierarchical clustering method	0.633	0.613	0.877	0.862

0.966, respectively. The precision and recall of the hierarchical clustering method were 0.877 and 0.862, respectively.

B. DBH ESTIMATION

To verify the effectiveness of the proposed DBH estimation method, we calculated the DBH values with the proposed DBH estimation method and circle fitting at 1.3 m height method, the point cloud circle fitting at 1.3 m height that is commonly used in the MLS point cloud DBH estimation [11], [14], [23], [24]. and all the trees we selected were measured by the DBH ruler to obtain the reference DBH value.

The DBH values of measurement and DBH estimation by this proposed method and comparison method are shown in Table 3, there are some trees were detected that the diameter at 1.3m height was an outlier of multi-height diameters, and the more accurate DBH values of these trees were obtained by the proposed DBH estimation method, there are five trees' DBH values were improved by the proposed method in the natural forest plot, and three trees' DBH values were improved by the proposed method in the urban forest plot.

The DBH values of measurement and DBH estimation by different methods are shown in Fig. 9, the DBH estimated

TABLE 3. The DBH values of the measured, this proposed method, and the comparison method (circle fitting at 1.3m).

Natural forest DBH (cm)			Urban forest DBH (cm)		
Measured (truth)	The proposed method	Comparison method	Measured (truth)	The proposed method	Comparison method
40.5	40.89	40.89	52.8	54.33	54.33
36.4	36.85	36.85	27.7	29.63	29.63
29.7	29.1	29.1	29.6	30.46	30.46
30.6	29.62	29.62	47.4	47.07	47.07
28.6	32.57	32.57	62.4	60.52	60.52
28.5	29.58	26.47	32.2	33.02	33.02
31.1	29.17	29.17	28.3	29.95	29.95
24.2	30.27	30.27	57.3	55.2	55.2
30.2	31.55	31.55	45.8	48.6	48.6
36.5	33.9	29.66	24.8	24.94	51.95
23.7	30.45	30.45	28.7	29.62	29.62
25.6	27.18	27.18	47.8	43.59	43.59
24.5	19.22	19.22	24.8	27.48	27.39
15.6	19.65	23.41	25.5	26.51	26.51
14.7	17.36	17.36	18.1	19.04	15.39
19.5	23.43	23.43	15.3	19.38	22.19
17.4	17.3	17.3	22.6	23.98	23.98
11.7	10.07	10.07	22.9	23.28	23.28
16.7	17.71	17.71	21.3	21.62	21.62
20.5	14.87	14.87	24.2	21.48	21.48
21.7	16.64	16.64	20.1	21.44	21.44
19.5	16.5	16.5	18.5	20.72	20.72
26.1	24.07	21.46	14.3	16.05	16.05
26.6	25.29	25.29			
27.5	27.41	27.41			
11.9	13.32	17.28			

by the proposed method are closer to the 1:1 line than the comparison method which is point cloud circle fitting at 1.3m height. It means that the proposed was more accurate than the comparison method.

The MAE, MAPE, and RMSE results were shown in Table 4, in the natural forest plot, the MAE of the proposed DBH estimation method was 2.5 cm, the MAPE was 11.54%, and the RMSE was 3.17 cm, while the MAE of the comparison method was 3.09 cm, the MAPE was 14.7%, and the RMSE was 3.88 cm. In the urban forest plot, the MAE of the proposed DBH estimation method was 1.65 cm, the MAPE was 6.31%, and the RMSE was 1.97 cm. The MAE of the comparison method was 3.02 cm, the MAPE was 12.27%, and the RMSE was 6.12 cm.

The DBH and distribution of trees were obtained. The DBH and relative positions of trees are shown in Fig. 10. Fig. 10 (a) is the DBH estimation results of the natural forest, and Fig. 10 (b) is the DBH estimation results of the urban forest.

IV. DISCUSSION

In this study, we attempted to segment the trees and estimate the DBH with backpack LiDAR data. For single tree segmentation, many previous works have been undertaken in pursuit of single tree segmentation. Bauwens *et al.* [32] segmented 91% of the trees with a hand-held mobile LiDAR in a mixed natural forest. Zhao *et al.* [33] detected the street trees with mobile laser scanning data, and the correctness rate of the segmentation results was over 92%. Wu *et al.* [34] detected the street trees with a voxel-based marked neighborhood searching method, and the completeness and correctness for tree detection were over 98%. Gollob *et al.* [35] used a new personal laser scanner to estimate the tree position, the detection rates were over 96% in different plots. Hyypä *et al.* [16] used the cross sections of the point clouds to segment the tree trunks and removed the noise points by a point density-based criterion, the completeness of stem detection reached 95.2% and 88.4% in different plots. Our proposed method used the relative density value to segment the

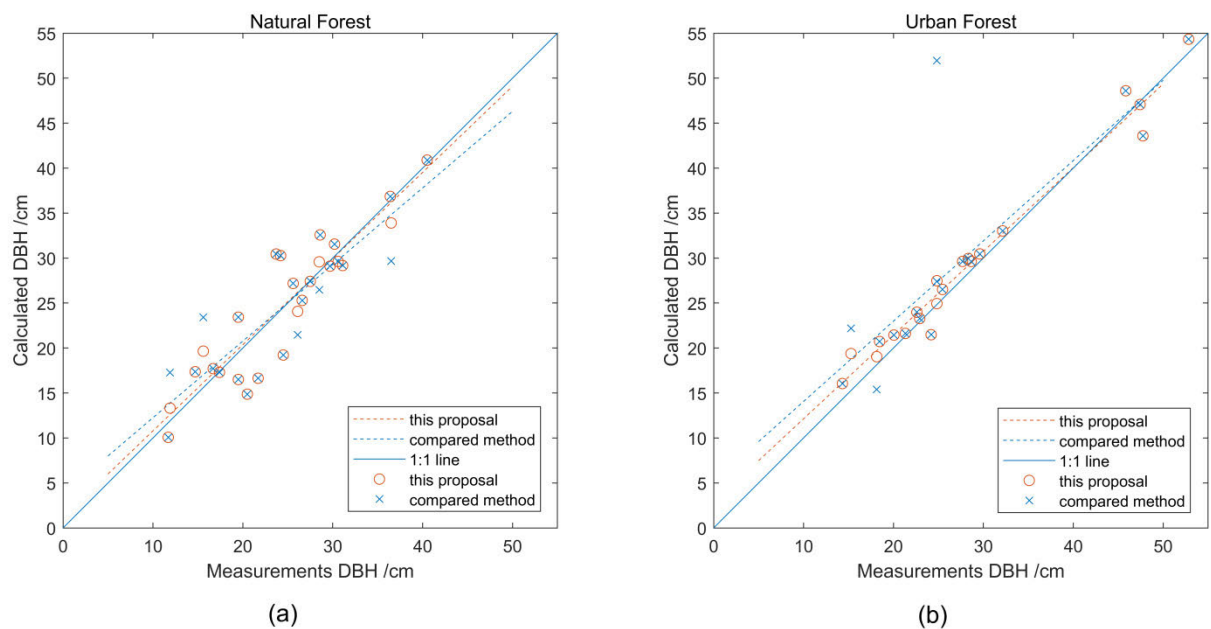


FIGURE 9. The DBH values of measurement and DBH estimation by different methods. (a) The DBH values in the natural forest plot, and (b) the DBH values in the urban forest plot.

TABLE 4. The results of MAE, MAPE, and RMSE in the DBH estimation of the proposed method and circle fitting at 1.3 m height.

Methods	Natural Forest			Urban Forest		
	MAE (cm)	MAPE (%)	RMSE (cm)	MAE (cm)	MAPE (%)	RMSE (cm)
Proposal method	2.5	11.54	3.17	1.65	6.31	1.97
Circle fitting at 1.3 m height	3.09	14.7	3.88	3.02	12.27	6.12

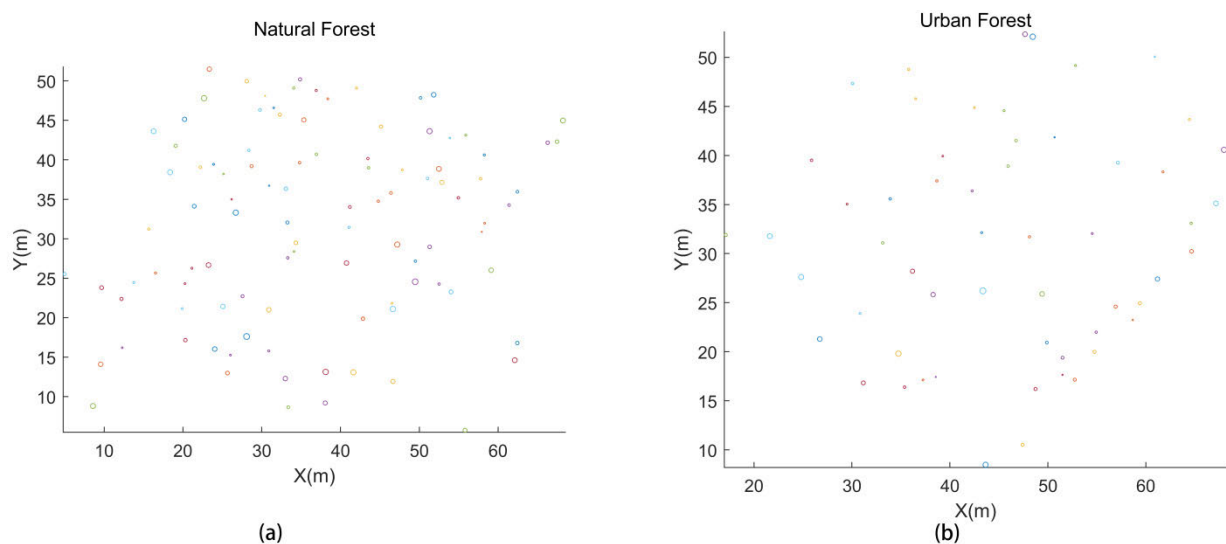


FIGURE 10. The DBH estimation results and relative position map of trees. (a) The results of trees in the natural forest, and (b) the results of trees in the urban forest.

tree trunks, it can avoid the MLS point cloud nonuniformity on density-based segmentation method. Meanwhile, the proposed method achieved single tree segmentation with the equivalent effect of other state-of-the-art methods as the precision and recall of the proposed method reached 0.966 and

0.946, respectively, in the complex natural forest and 1.00 and 0.966, respectively, in the easy urban forest. Compared with the results of the comparison method [13], [14], our proposed method obtained a better result in the same forest environment.

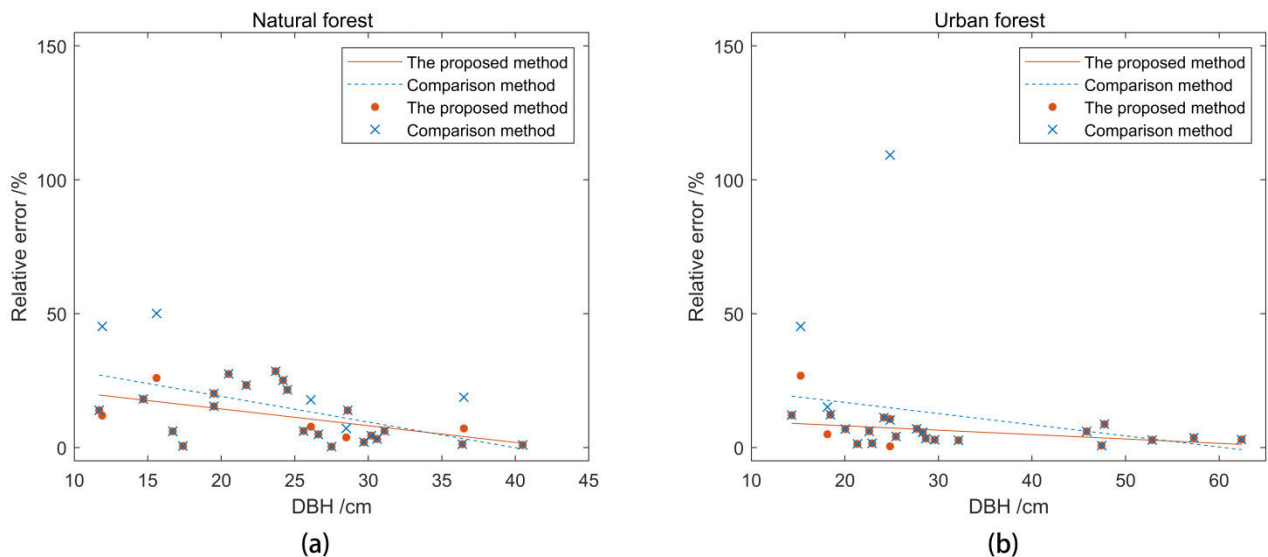


FIGURE 11. The relationship of the relative error with the measured DBH. (a) The relationship of the relative error with the measured DBH in the natural forest plot, and (b) the relationship of the relative error with the measured DBH in the urban forest plot.

For the DBH estimation with mobile laser scanning data, Gollob *et al.* [35] estimated the DBH with the circular cluster method and root mean square error (RMSE) of the best DBH measurement was 2.32 cm. Chen *et al.* [36] estimated the DBH with the personal laser scanning (PLS) system, and the RMSE was 1.58 cm in an arbor forest sample plot. Perugia *et al.* [37] estimated the DBH on different types of forest, and the RMSE was 2.45–4.78 cm depending on the different forest types. The proposed DBH estimation method can achieve an accuracy comparable to other state-of-the-art methods as the RMSE of the proposed method was 3.31 cm in the natural forest plot and 1.97 cm in the urban forest plot. Meanwhile, the DBH values obtained by the proposed method show a strong correlation with the measured values, compared with the comparison method in the same forest plot, the proposed method provided a better DBH estimation result.

Analysis of the connection between the error and DBH revealed that the error of DBH estimation decreased while the measured trees DBH increasing, as shown in Fig. 11. The proposed method performed better when the DBH was small compared with the comparison method. As the clear bole height (CBH) of trees were related to the DBH [38], the trees with smaller DBH always have smaller CBH, the DBH estimation was easily affected by the extra branch point cloud. Meanwhile, every point obtained by LiDAR has a certain distance to every other point, and the distance between points increases with the distance between the target and LiDAR. The smaller the DBH of the tree, the fewer points the LiDAR can be acquired, and thus it is difficult to fit the tree shape accurately with the points, therefore, the trees with smaller DBH were generally difficult to estimate the tree's DBH accurately. The proposed method used the multi-height diameters to improve the accuracy and robustness of DBH estimation, which can maintain high accuracy with poor trunk point cloud quality.

In this study, we provide an accurate and robust DBH extraction method for the forest backpack LiDAR data, which maintains high accuracy despite low trunk point cloud data quality and provides more reliable data for forest resource surveys. However, there are some inadequacies with the proposed segmentation method, the parameters of the proposed segmentation method need to be set manually to get the best results, and the parameters may change based on the different forest environment or MLS point cloud data, therefore, it is necessary to find an adaptive parameters method for the different forest type or MLS point cloud data in future work. Meanwhile, the total area of the experimental data is about 5600m², it would be more convincing if the proposed method is verified in a larger area, we will collect a larger area data in the future work.

V. CONCLUSION

With the wide application of mobile LiDAR scanning in forestry, obtaining the 3D information of trees is becoming more convenient. In this study, we proposed a single tree segmentation and DBH estimation method based on a backpack mobile LiDAR system, and the accuracy and robustness of the proposed method were verified in both natural forest and urban forest conditions. For the single tree point cloud segmentation, this study proposed a relative density segmentation method, which effectively reduced the influence of shrubs and branches on forest point cloud segmentation, and improved the point cloud segmentation accuracy. For the DBH estimation, we proposed a DBH estimation method based on multi-height diameters, which could still obtain relatively accurate estimation results when the data quality of the point cloud is poor.

In the complex natural forest environment, the precision of the proposed segmentation method reached 0.966 and the recall reached 0.946. In the urban forest environment, the precision of the proposed segmentation method reached 1.0 and

the recall reached 0.966. Comparing the MAE, MAPE, and RMSE results, the proposed DBH estimation method result performed better than the comparison method in the same forest environment. The results of this study demonstrated the high accuracy and robustness of the proposed method and the potential of the backpack mobile LiDAR system for forest inventory in a complex environment.

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